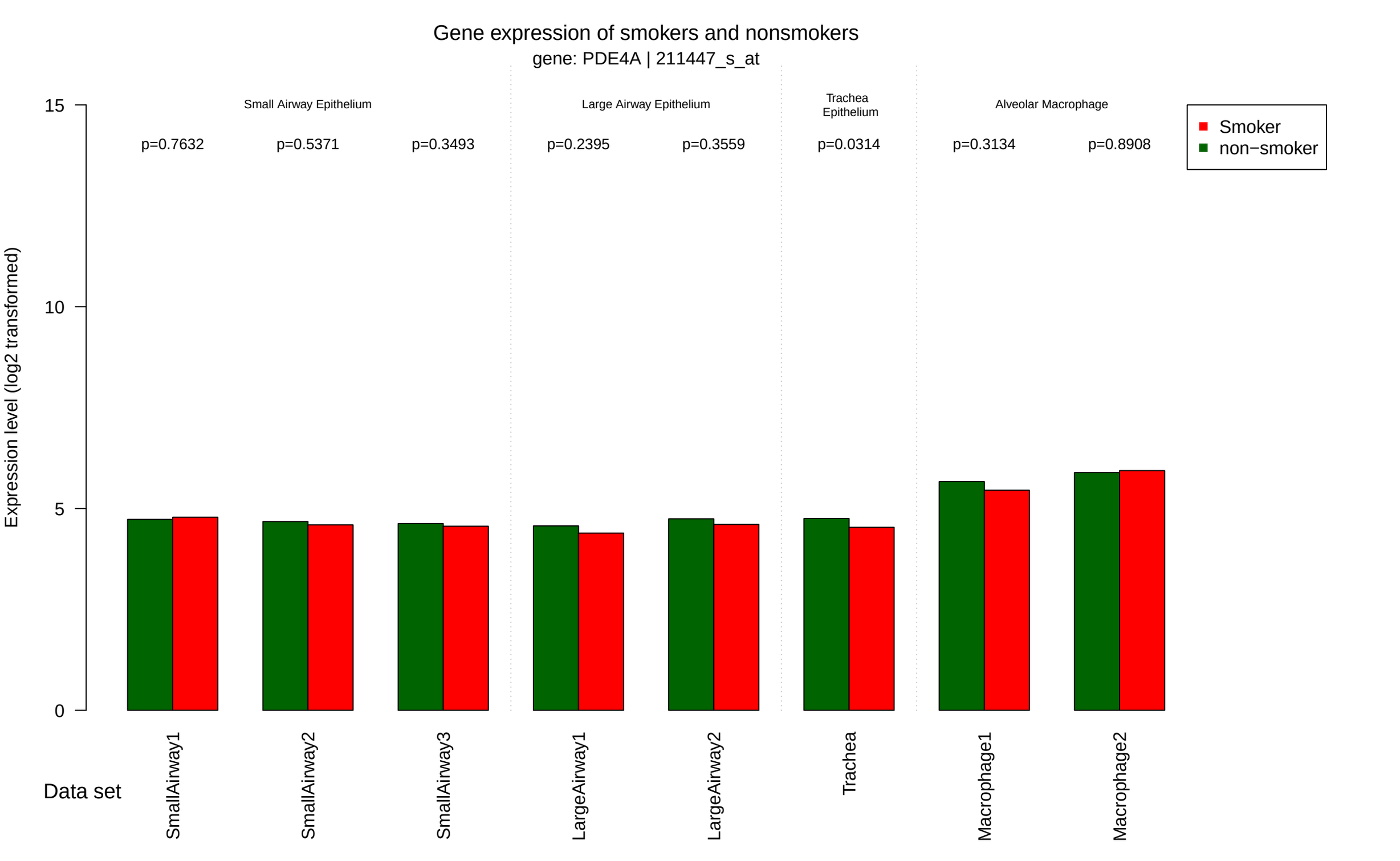
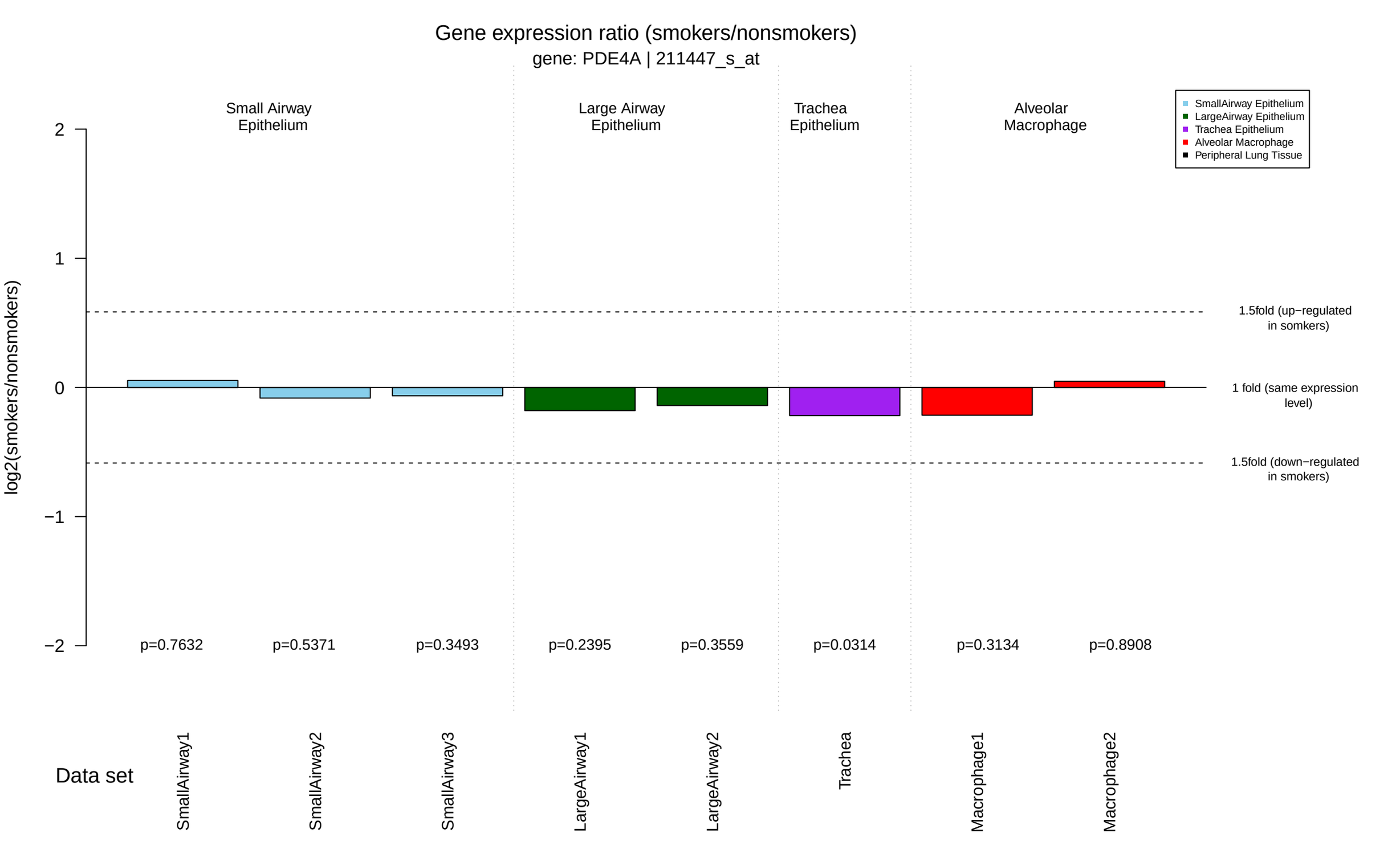
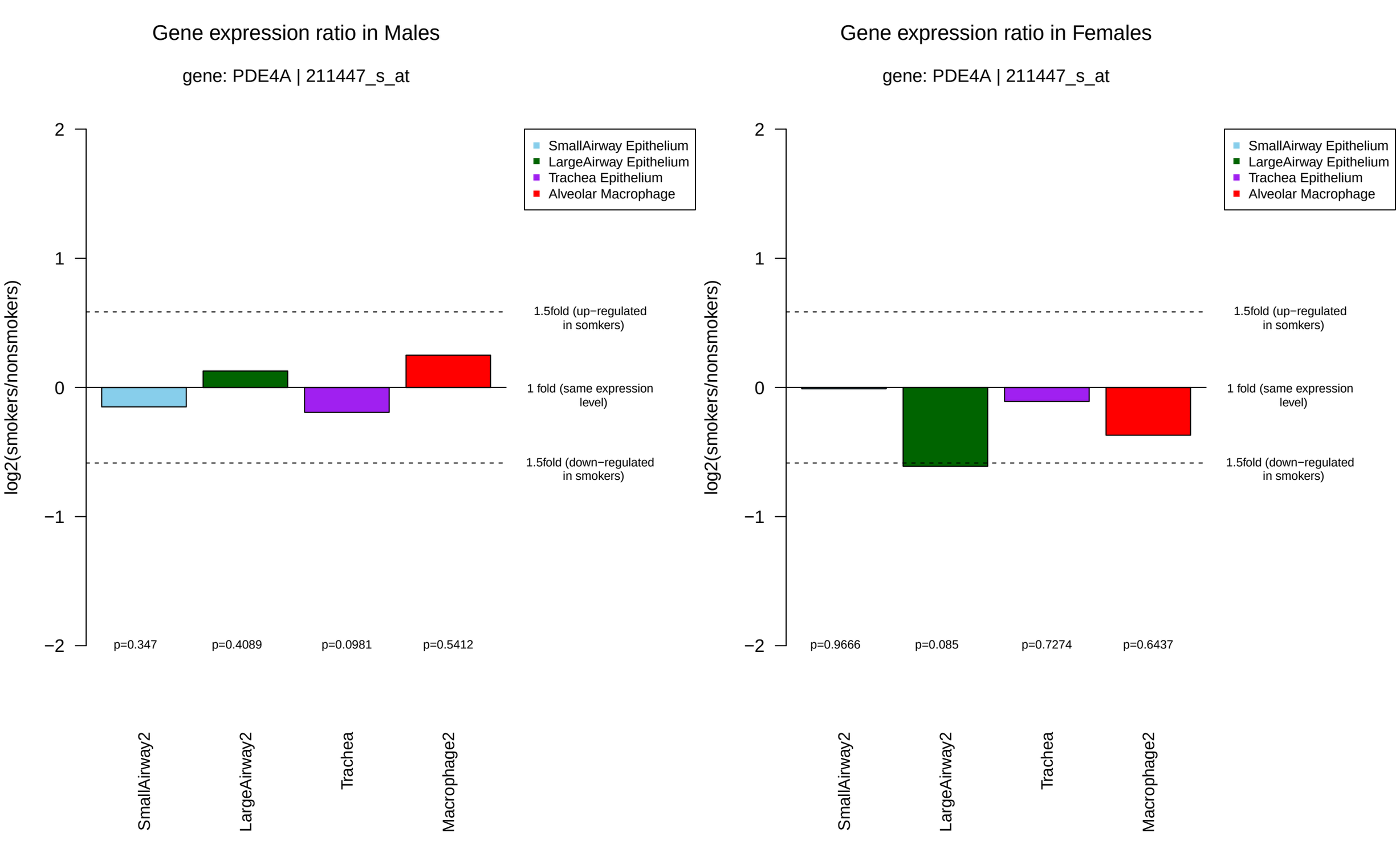
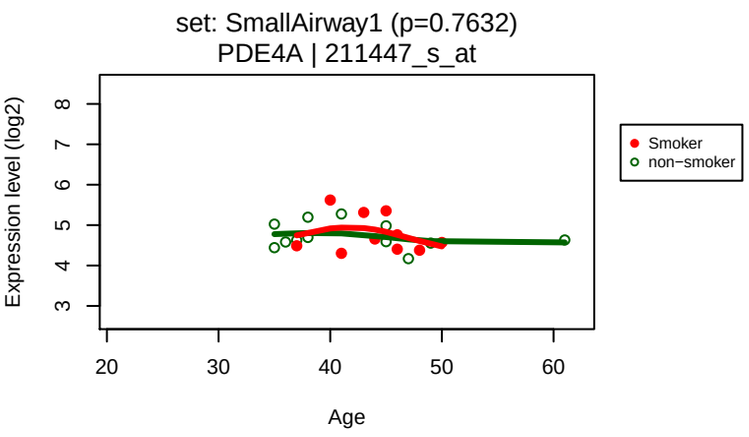
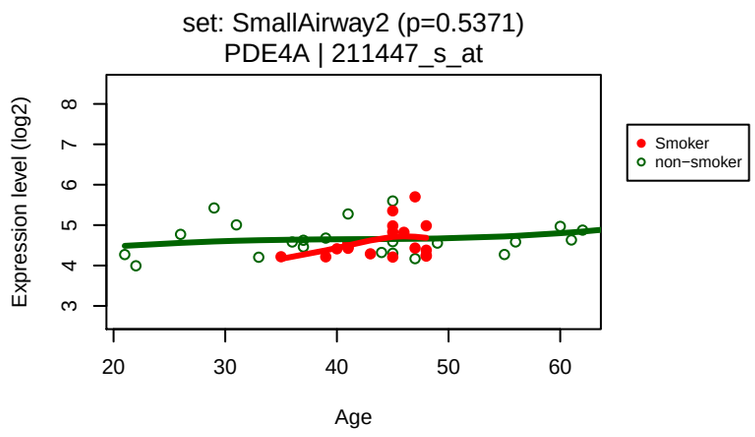
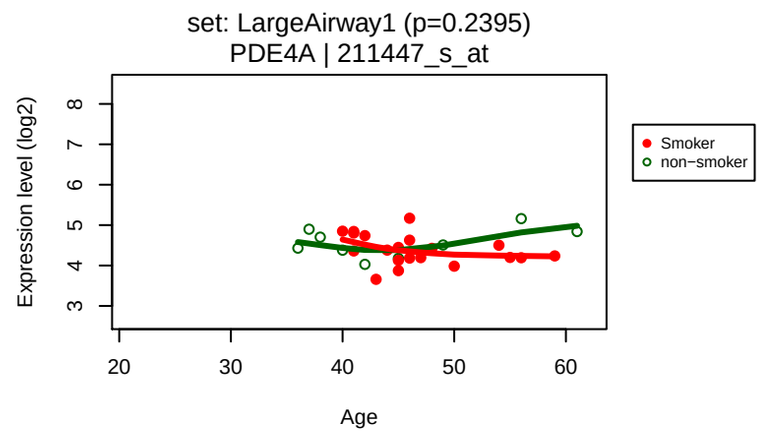

Gene expression of smokers and nonsmokers
gene: PDE4A | 211447_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7632
p=0.5371
p=0.3493
p=0.2395
p=0.3559
p=0.0314
p=0.3134
p=0.8908
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

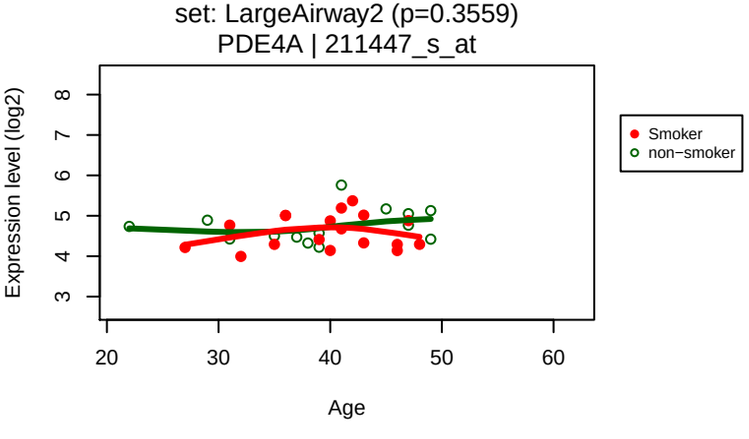

Gene expression ratio (smokers/nonsmokers)
gene: PDE4A | 211447_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7632
p=0.5371
p=0.3493
p=0.2395
p=0.3559
p=0.0314
p=0.3134
p=0.8908
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

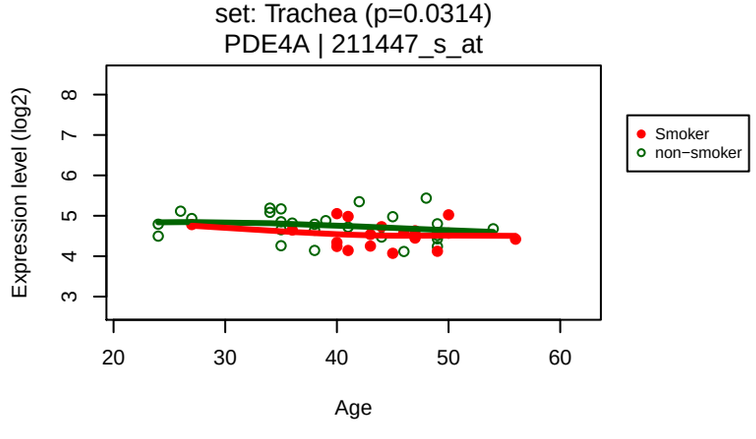

Gene expression ratio in Males
gene: PDE4A | 211447_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.347
p=0.4089
p=0.0981
p=0.5412
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE4A | 211447_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9666
p=0.085
p=0.7274
p=0.6437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

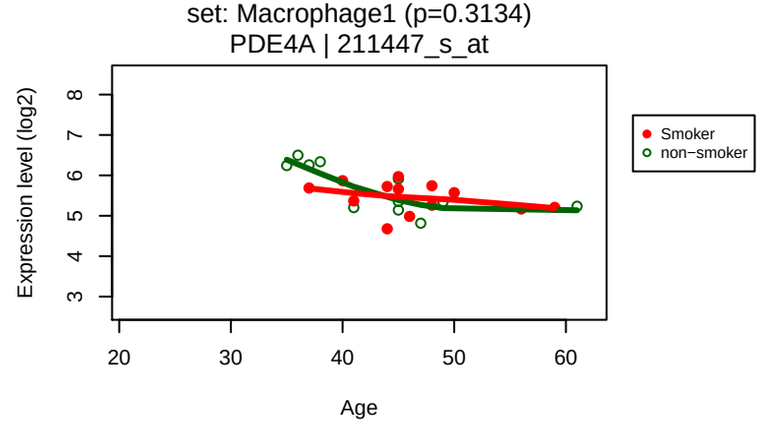

set: SmallAirway1 (p=0.7632)
PDE4A | 211447_s_at
Expression level (log2)
Age
Smoker
non−smoker

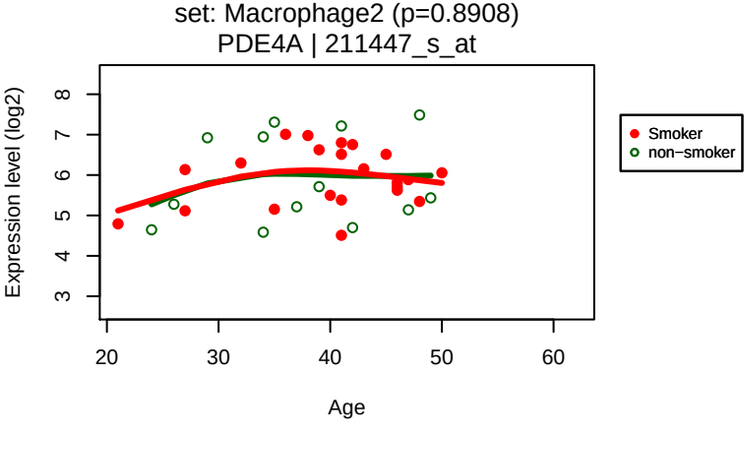

set: SmallAirway2 (p=0.5371)
PDE4A | 211447_s_at
Expression level (log2)
Age
Smoker
non−smoker

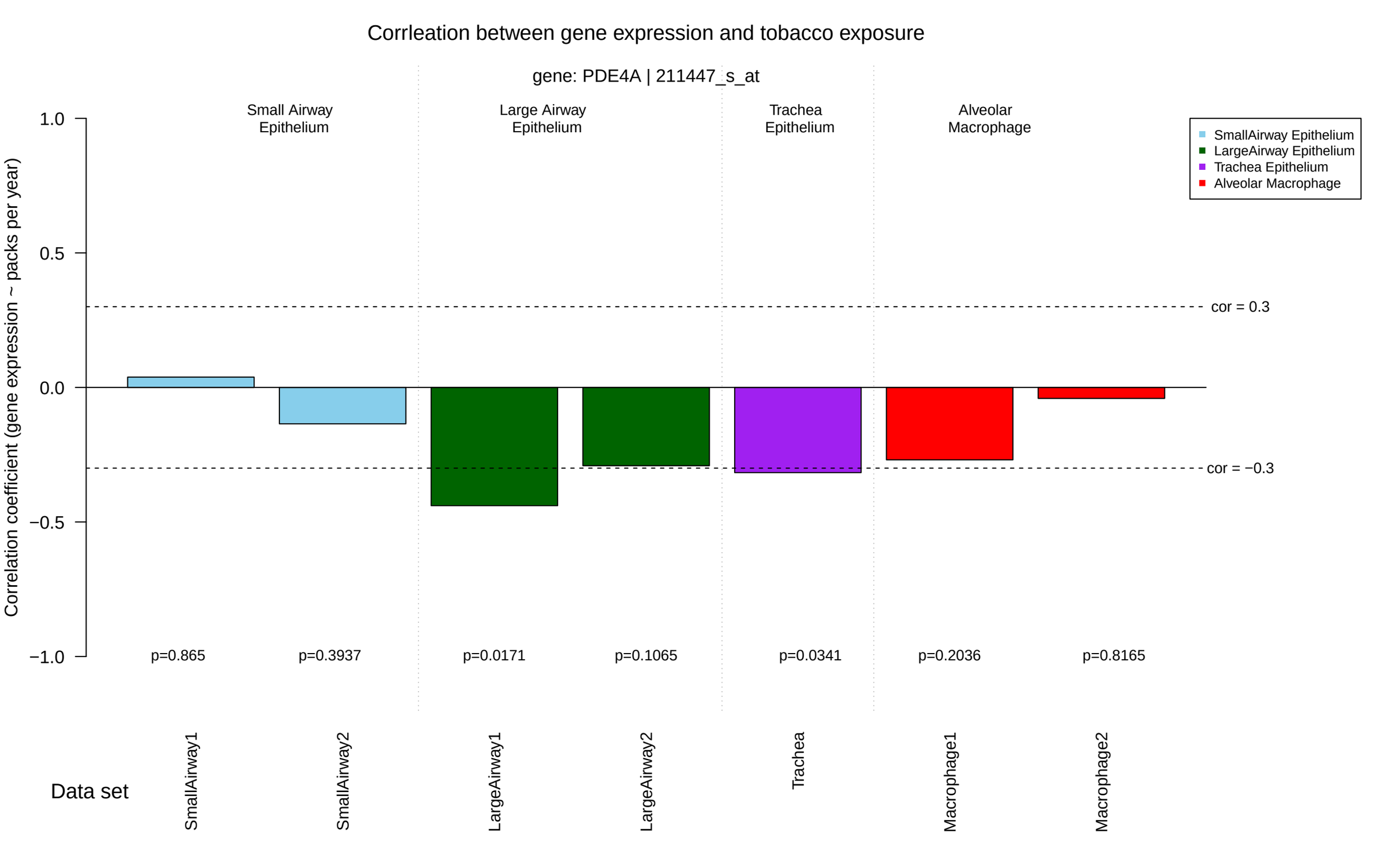

set: LargeAirway1 (p=0.2395)
PDE4A | 211447_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3559)
PDE4A | 211447_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0314)
PDE4A | 211447_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3134)
PDE4A | 211447_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8908)
PDE4A | 211447_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE4A | 211447_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.865
p=0.3937
p=0.0171
p=0.1065
p=0.0341
p=0.2036
p=0.8165
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2